

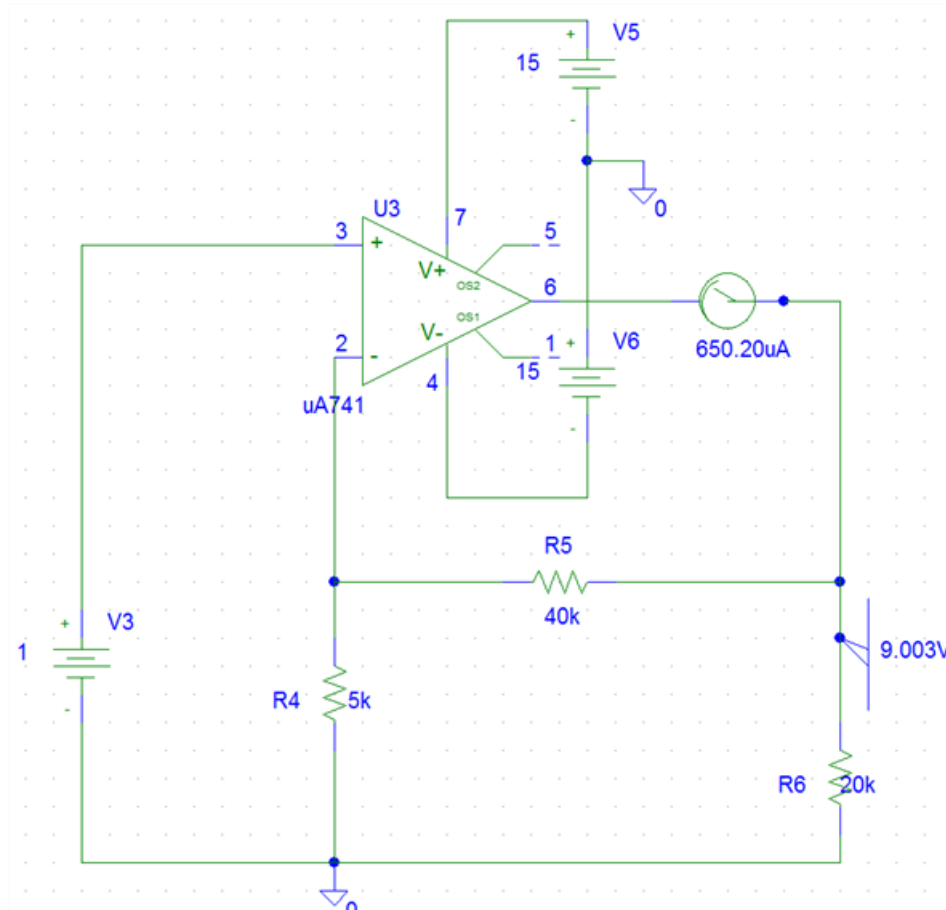
Problem

Rework Practice Prob. 5.1 using *PSpice*.

Step-by-step solution

Step 1 of 2

The first step is to draw the given circuit using schematics



Step 2 of 2

The positive terminal of the voltage source v_s is connected to the inverting terminal Pin 2 with the 5 k resistor

The non inverting terminal P in 3 is connected with 1 V

As the OP amp is powered

The positive power supply terminal $V^+(P\text{ in }7)$ is connected to 15 V dc voltage source

Where the negative power supply terminal $V^-(P\text{ in }4)$ is connected to -15 V

The P in 1 and 5 are left floating because they are used for offset null adjustment

Besides adding the dc power supplies to the original circuit

We have also added pseudo components view point and lprobe measure the O/P voltage and currents respectively.

After saving the schematic we simulated the circuit by selections Analysis/ Simulate and have the results displayed on view point and lprobe. From the circuit results we have

$$i = 650.2\text{ }\mu\text{A}$$

$$i = 0.6502\text{ mA}$$

$$V = 9.003\text{ V}$$